Putnam

Recurrences

Written by a model for Adit. Not Adit's words.

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1 Recurrences

Recognise it. A sequence is defined from its earlier terms and something is asked about a far term, about divisibility of the terms, about their growth or their limit. A count for n splits according to what happens first or last, which builds it from counts for smaller n. A sequence of polynomials, matrices or functions is defined by iteration. A problem about "the 2026th term" is a problem about the period. A rational recurrence whose terms stay integers, or a nonlinear one whose terms stay bounded, is almost always a linear one in disguise.

Definition 1.1. A linear recurrence of order k with constant coefficients is $a_{n+k} = c_{k-1}a_{n+k-1} + \cdots + c_0 a_n$ for all $n \geq 0$, with $c_0 \neq 0$. Its characteristic polynomial is $p(x) = x^k - c_{k-1}x^{k-1} - \cdots - c_0$. Writing E for the shift $(Ea)_n = a_{n+1}$, the recurrence says $p(E)a = 0$.

Theorem 1.2 (Solving a linear recurrence)**.** *The sequences satisfying a linear recurrence of order k form a vector space V of dimension k. If $p(x) = \prod_{i=1}^{s}(x - r_i)^{m_i}$ with distinct $r_i \neq 0$, then the k sequences $n^j r_i^n$, $0 \leq j < m_i$, form a basis of V. So every solution is*

$$a_n = \sum_{i=1}^{s} P_i(n)\, r_i^n, \qquad \deg P_i < m_i,$$

and when all roots are simple, $a_n = \sum_i C_i r_i^n$ with constants C_i fixed by the first k terms.

Proof. A solution is determined by $(a_0, \ldots, a_{k-1})$ and every choice of those values gives a solution, so the map $V \to \mathbb{R}^k$ (or $\mathbb{C}^k$) sending a solution to its first k terms is a linear bijection and $\dim V = k$.

For a single factor: $(E-r)$ applied to $n^j r^n$ gives $(n+1)^j r^{n+1} - r \cdot n^j r^n = r^{n+1}\big((n+1)^j - n^j\big)$, and $(n+1)^j - n^j$ is a polynomial in n of degree $j-1$ (zero when $j = 0$). So each application of $(E-r)$ lowers the polynomial degree by one, and $(E-r)^m$ kills $n^j r^n$ for every $j < m$. The factors $(E - r_i)^{m_i}$ commute, so $p(E)$ kills all k listed sequences and they lie in V.

Linear independence: suppose $\sum_i P_i(n) r_i^n = 0$ for all n with some $P_i \neq 0$. Order the roots with $|r_1| \geq |r_2| \geq \ldots$; among the roots of largest modulus and largest $\deg P_i$,

divide by $n^d r_1^n$ where d is that degree. Terms with smaller modulus or smaller degree tend to 0, and what remains is a finite sum of the form $\sum_j c_j\, e^{in\theta_j}$ with distinct θ_j and not all c_j zero, which does not tend to 0 (its average over n of $|\cdot|^2$ is $\sum |c_j|^2 > 0$). Contradiction. Hence the k sequences are independent and, by dimension, a basis. □

A check worth doing once by hand: $a_{n+2} = 4a_{n+1} - 4a_n$ has $p(x) = (x-2)^2$, so $a_n = (\alpha + \beta n)2^n$; with $a_0 = 1$, $a_1 = 4$ this is $(1+n)2^n$, and indeed $a_2 = 4\cdot 4 - 4\cdot 1 = 12 = 3\cdot 4$. Forgetting the factor n for the repeated root is the commonest error in this section.

Proposition 1.3 (Eventual periodicity modulo m)**.** *Let $m \geq 2$ and let a sequence satisfy*

$$a_{n+k} = f(a_n, \ldots, a_{n+k-1}),$$

where f maps $(\mathbb{Z}/m)^k$ to $\mathbb{Z}/m$. Then the sequence $(a_n \bmod m)$ is eventually periodic with period at most m^k. If moreover f is a bijection in its first argument for every fixed value of the others (for a linear recurrence, when c_0 is a unit modulo m), the sequence is periodic from the start.

Proof. The state $s_n = (a_n, \ldots, a_{n+k-1}) \bmod m$ determines all later states, and there are only m^k states, so two of $s_0, \ldots, s_{m^k}$ coincide, say $s_i = s_j$ with $i < j$; then $s_{n+(j-i)} = s_n$ for all $n \geq i$. When f is invertible in its first argument, s_n is recovered from s_{n+1} as well, so the map on states is a bijection of a finite set and every orbit is a cycle: $s_i = s_j$ implies $s_0 = s_{j-i}$. □

Proposition 1.4 (Linearisation by substitution)**.** *Each of the following nonlinear recurrences is a linear one after a change of variable.*

1. $x_{n+1} = \dfrac{\alpha x_n + \beta}{\gamma x_n + \delta}$ *with* $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ *invertible. If* $(p_{n+1}, q_{n+1})^T = M(p_n, q_n)^T$ *and* $x_0 = p_0/q_0$*, then* $x_n = p_n/q_n$ *whenever* $q_n \neq 0$*. So x_n is a ratio of two sequences satisfying the linear recurrence with characteristic polynomial* $\det(xI - M)$*, and M^k scalar forces period k.*
2. $x_{n+1} = x_n^2 - 2$*: with* $x_n = t_n + t_n^{-1}$ *one gets* $t_{n+1} = t_n^2$*, so* $x_n = t_0^{2^n} + t_0^{-2^n}$*; this needs* $|x_0| \geq 2$ *for real t_0, else* $x_0 = 2\cos\theta$ *and* $x_n = 2\cos(2^n\theta)$*.*
3. $x_{n+1} = \dfrac{x_n + c}{1 - cx_n}$*: with* $x_n = \tan\theta_n$ *and* $c = \tan\varphi$ *one gets* $\theta_{n+1} = \theta_n + \varphi$*.*
4. $x_{n+1} = x_n^r \cdot K$*: take logarithms,* $\ln x_{n+1} = r \ln x_n + \ln K$*, a first-order linear recurrence.*

Proof. (1) $\dfrac{p_{n+1}}{q_{n+1}} = \dfrac{\alpha p_n + \beta q_n}{\gamma p_n + \delta q_n} = \dfrac{\alpha x_n + \beta}{\gamma x_n + \delta}$ after dividing by q_n. By Cayley-Hamilton both p_n and q_n satisfy $u_{n+2} = \operatorname{tr}(M)u_{n+1} - \det(M)u_n$. (2) $(t + t^{-1})^2 - 2 = t^2 + t^{-2}$. (3) is the tangent addition formula. (4) is immediate. □

The other two pieces of the model come from neighbouring sections. A closed form guessed from the first terms is proved by induction, and the induction step is usually just the recurrence. A count c_n that splits by the first object satisfies $c_n = \sum_j (\text{ways the first part is } j) \cdot c_{n-j}$, which is the convolution shape that generating functions handle.

First moves. Compute six terms and look at them: Fibonacci, Lucas, Catalan, powers of 2, factorials, triangular numbers and their small relatives are the usual disguises. Compute the first differences and the ratios of consecutive terms. Compute

modulo 2, 3, 4 and 5 and look for the period. For a linear recurrence write the characteristic polynomial and factor it; a repeated root or a root of unity is a flag. For a rational or square-root recurrence try the substitutions above, in the order ratio, tangent, $t + 1/t$, logarithm. For a recurrence with polynomial coefficients, write the generating function.

Example 1.5. Define $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = 6a_{n+1} - a_n$ for $n \geq 0$. Find a closed form for a_n, and show that 7 divides a_n if and only if 3 divides n.

Solution. The characteristic polynomial is $x^2 - 6x + 1$ with roots $r = 3 + 2\sqrt{2}$ and $\bar{r} = 3 - 2\sqrt{2}$, distinct, so $a_n = Cr^n + D\bar{r}^{\,n}$. From $a_0 = 0$, $D = -C$; from $a_1 = 1$, $C(r - \bar{r}) = 4\sqrt{2}\,C = 1$. Hence

$$a_n = \frac{(3 + 2\sqrt{2})^n - (3 - 2\sqrt{2})^n}{4\sqrt{2}}.$$

The first terms are $0, 1, 6, 35, 204, 1189, 6930$, and $35 = 5 \cdot 7$, $6930 = 7 \cdot 990$, while the others are not multiples of 7. This suggests the claim.

Modulo 7 the coefficient of a_n in the recurrence is -1, a unit, so the residues are purely periodic by the proposition. Compute: $a_0 \equiv 0$, $a_1 \equiv 1$, $a_2 \equiv 6$, $a_3 \equiv 36 - 1 = 35 \equiv 0$, $a_4 \equiv 0 - 6 \equiv 1$, $a_5 \equiv 6 - 0 = 6$, and the state $(a_3, a_4) \equiv (0, 1) = (a_0, a_1)$ has returned. So the residues repeat with period 3: $a_n \equiv 0, 1, 6$ according as $n \equiv 0, 1, 2 \pmod 3$, and $7 \mid a_n$ exactly when $3 \mid n$.

Two remarks. The closed form was not needed for the divisibility, and could not have given it without more work; the finite state space did. And the proof of periodicity is the two lines of computation plus the sentence saying why a returning state closes the cycle, which is the sentence graders look for. □

Example 1.6. Let $x_0 = 2$ and $x_{n+1} = \dfrac{x_n + 1}{1 - x_n}$ for $n \geq 0$. Show that x_n is defined for every n, and find x_{2026}.

Solution. The recurrence is the Möbius map with matrix $M = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. Compute $M^2 = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ and $M^4 = (M^2)^2 = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix} = -4I$, a scalar. So if $(p_n, q_n)^T = M^n(2, 1)^T$, then $(p_{n+4}, q_{n+4}) = -4(p_n, q_n)$ and $x_{n+4} = x_n$ whenever the quotients are defined.

Directly: $x_0 = 2$, $x_1 = 3/(-1) = -3$, $x_2 = (-2)/4 = -\frac{1}{2}$, $x_3 = (\frac{1}{2})/(\frac{3}{2}) = \frac{1}{3}$, $x_4 = (\frac{4}{3})/(\frac{2}{3}) = 2 = x_0$. None of these four values is 1, so the denominator never vanishes, and the sequence is periodic with period 4 from the start. Since $2026 = 4 \cdot 506 + 2$, $x_{2026} = x_2 = -\frac{1}{2}$.

The substitution view says the same thing: with $x_n = \tan\theta_n$ the map is $\theta_{n+1} = \theta_n + \pi/4$, and four steps add π, which does not change the tangent. Either view is a complete proof once the four values are listed; the matrix view also explains why the period is 4 before any term is computed, since M is $\sqrt{2}$ times a rotation by $45°$. □

Trap. A repeated characteristic root, which needs the factor n; check by computing one more term than the number of constants. Initial terms that break the pattern: a recurrence valid for $n \geq 2$ with a_0, a_1 not fitting the closed form, so the closed form must be stated for $n \geq 2$ and the early terms listed separately. Assuming periodicity starts at the first term when the map on states is not a bijection; $a_{n+1} = a_n^2 \bmod 10$ from $a_0 = 2$ runs $2, 4, 6, 6, 6, \ldots$ and is eventually constant, not periodic from the start. A substitution that is only valid on part of the domain, like $t + 1/t$ for $|x| \geq 2$. Reporting a period found numerically without the sentence that a repeated state forces it. A

limit of a recurrence computed by solving $L = f(L)$ before the sequence has been shown to converge.

Ladder.

1. 2016 B1. A sequence defined through logarithms; sum a series built from it.
2. PS 4 #1. A third-order linear recurrence; the terms as a sum of familiar sequences.
3. PS 4 #3. The parity of the integer part of a power; a recurrence hidden in a surd.
4. PS 4 #5. The ceiling of a power of a surd; divisibility by a power of two.
5. PS 4 #9. A second-order recurrence; one specific far term.
6. 2017 A2. A rational recurrence whose terms turn out to be integer polynomials.
7. 2025 A1. An iteration on fractions; the terms eventually coprime.
8. 2020 A3. An iterated sine; convergence of a series of the terms.
9. 2018 B4. A cosine-like recurrence; one zero term forces periodicity.
10. 2021 B4. A product of powers reduced modulo a Fibonacci number; the remainder.
11. 2019 B5. A polynomial interpolating a famous recurrence; extrapolate one step.

Where it hides. Inside counting, where the count for n splits by the first part (2020 A5, 2022 A3, 2024 B5). Inside probability, where conditioning on the first step turns a probability or an expectation into a recurrence (2020 A4, 2020 B3, 2024 B4). Inside number theory, where the exact power of a prime dividing a term follows the recurrence (2025 A6, 2021 B4). Inside polynomials, where a sequence of polynomials is defined by iteration (2017 A2, 2019 B5). Inside linear algebra, where the recurrence is a power of a matrix (PS 7 #24, PS 7 #25).